

MTH208: Worksheet 16

More Rcpp

Using the `cppFunction()` to integrate C++ code in R, is very useful. However, it does not provide easy editing and indenting. Further, it is not practical to write big C++ codes in this way. A solution to this is `sourceCpp()` function.

Click on a New C++ file button, which creates a new file with C++ code that Rcpp can read. Save this file **in your working directory**.

The first two lines of the C++ file will be

```
#include <Rcpp.h>
using namespace Rcpp;
```

Note that the hashtag before include is necessary. Since this is a C++ file, `#` is not a comment. Comments in C++ are using `//`.

Then, right before the function starts, you must write the following comment line to let Rcpp know that this is the main function:

```
// [[Rcpp::export]]
NumericVector dummy(NumericVector x){
  return x;
}
```

Once your file is written, in a separate R file, you can compile the `*.cpp` file using the `sourceCpp("*.cpp")` (this works because the file is in your working directory)

```
library(Rcpp)
sourceCpp("*.cpp") # replace * with name of file
```

Finally, you can learn about the various functions in Rcpp here: https://teuder.github.io/rcpp4everyone_en/index.html.

For generic C++ code see here: <https://www.w3schools.com/cpp/default.asp>

Problems

1. Redo Worksheet 15 problems using `sourceCpp()`.
2. Using Section~22 in the given Rcpp reference, redo Problem 3 in Worksheet 3. Benchmark with the

solution using R.

The format for the while loop is:

```
int i;
while(i < 10){
    ...
    i++;
}
```

3. Using Section~22 in the given Rcpp reference, redo Problem 4 in Worksheet 3. Benchmark with the solution using R.
4. As you can see while dealing with matrices in C++, it can be annoying to write loops for elementary tasks like adding matrices. The `RcppArmadillo` library in R provides matrix compatibility.

However, the namespace changes: `NumericVector` becomes `vec`, `NumericMatrix` becomes `mat`. For example, finding the inner product of two vectors, the cpp file looks like

```
// [[Rcpp::depends(RcppArmadillo)]]
#include <RcppArmadillo.h>
using namespace Rcpp;
using namespace arma;

// [[Rcpp::export]]
double timesTwo(arma::vec x, arma::vec y) {
    // inner product of two vectors is a matrix
    mat inn(1,1);
    inn = x.t() * y;

    // returning the first element of a 1x1 matrix
    return inn(0);
}
```

Using what you've learned here, write a C++ code using `RcppArmadillo` that adds two matrices and benchmark with the previous worksheet.

More armadillo operations are here: <https://arma.sourceforge.net/docs.html>.